

KRUTIK

CYBER EXPERT

CSS COMPLETE COURSE

Styling + Layout + Animations

BY KRUTIK 
CYBER EXPERT

"Learn Free. Design Secure."

CHAPTER 1: CSS INTRODUCTION

What is CSS?

- Cascading Style Sheets
- Created by Håkon Wium Lie (1996)
- Latest Version: CSS3
- Used for styling and layout of web pages

Why CSS?

- ✓ Makes websites beautiful
- ✓ Separates content (HTML) from design (CSS)
- ✓ Responsive design (mobile friendly)
- ✓ Reusable styles (save time)
- ✓ Animations and effects
- ✓ All browsers support

What can you do with CSS?

- ✓ Colors, fonts, backgrounds
- ✓ Layouts (Flexbox, Grid)
- ✓ Animations & Transitions

- ✓ Responsive Design
- ✓ Print Styles
- ✓ Dark/Light Mode

CHAPTER 2: HOW TO ADD CSS (3 Ways)

A) INLINE CSS (Inside HTML Tag):

CODE:

```
<h1 style="color: blue; font-size: 30px;">Hello World</h1>
<p style="background-color: yellow; padding: 10px;">Styled paragraph</p>
```

B) INTERNAL CSS (Inside <head>):

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    h1 {
      color: blue;
      font-size: 30px;
    }
    p {
      background-color: yellow;
      padding: 10px;
    }
  </style>
</head>
<body>
  <h1>Hello World</h1>
  <p>Styled paragraph</p>
</body>
</html>
```

C) EXTERNAL CSS (Separate .css file):

CODE (style.css file):

```
h1 {  
  color: blue;  
  font-size: 30px;  
}  
p {  
  background-color: yellow;  
  padding: 10px;  
}
```

CODE (HTML file):

```
<!DOCTYPE html>  
<html>  
<head>  
  <link rel="stylesheet" href="style.css">  
</head>  
<body>  
  <h1>Hello World</h1>  
  <p>Styled paragraph</p>  
</body>  
</html>
```

D) PRIORITY ORDER (Which wins?):

1. Inline CSS (Highest Priority) - `style=""`
2. Internal CSS - `<style>` tag
3. External CSS - `.css` file
4. Browser Default (Lowest Priority)

RULE: More specific = Higher priority

CHAPTER 3: CSS SELECTORS (Target Elements)

A) BASIC SELECTORS:

CODE:

/* 1. Universal Selector - selects everything */

```
* {  
  margin: 0;  
  padding: 0;  
}
```

/* 2. Element Selector - selects by tag name */

```
p {  
  color: red;  
}  
h1 {  
  font-size: 30px;  
}
```

/* 3. Class Selector - selects by class (.) */

```
.highlight {  
  background-color: yellow;  
}  
.text-large {  
  font-size: 24px;  
}
```

/* 4. ID Selector - selects by id (#) */

```
#header {  
  background-color: blue;  
  color: white;  
}  
#logo {  
  width: 100px;  
}
```

/* 5. Group Selector - multiple selectors */

```
h1, h2, h3 {  
  color: navy;  
}
```

/* 6. Descendant Selector - space */

```
div p {  
  color: green;  
}  
.container .box {  
  border: 1px solid black;  
}
```

/* 7. Child Selector - > (direct child) */

```
div > p {  
  font-weight: bold;  
}
```

```
/* 8. Adjacent Sibling - + (next sibling) */
h1 + p {
    margin-top: 0;
}
```

```
/* 9. General Sibling - ~ (all siblings) */
h1 ~ p {
    color: gray;
}
```

B) ATTRIBUTE SELECTORS:

CODE:

```
/* Elements with specific attribute */
input[type="text"] {
    border: 2px solid blue;
}
input[type="submit"] {
    background-color: green;
    color: white;
}
```

```
/* Starts with */
a[href^="https"] {
    color: green;
}
```

```
/* Ends with */
a[href$=".pdf"] {
    color: red;
}
```

```
/* Contains */
a[href*="google"] {
    color: orange;
}
```

C) PSEUDO-CLASSES (State-based):

CODE:

```
/* Link states */
```

```
a:link { color: blue; }    /* Unvisited link */
a:visited { color: purple; } /* Visited link */
a:hover { color: red; }    /* Mouse over */
a:active { color: green; } /* Clicking */
```

```
/* Hover on any element */
button:hover {
    background-color: darkblue;
}
div:hover {
    transform: scale(1.05);
}
```

```
/* Focus (input fields) */
input:focus {
    border-color: blue;
    outline: none;
}
```

```
/* First/last child */
p:first-child {
    color: red;
}
p:last-child {
    color: blue;
}
```

```
/* nth-child */
li:nth-child(2) { color: red; }    /* 2nd item */
li:nth-child(even) { background: #f0f0f0; } /* Even items */
li:nth-child(odd) { background: white; }    /* Odd items */
li:nth-child(3n) { color: green; } /* Every 3rd item */
li:nth-child(3n+1) { color: blue; } /* 1st, 4th, 7th... */
```

D) PSEUDO-ELEMENTS (::):

CODE:

```
/* First line */
p::first-line {
    font-weight: bold;
    color: red;
}
```

```
/* First letter */
p::first-letter {
```

```

    font-size: 40px;
    color: blue;
}

/* Before (insert content before) */
h1::before {
    content: "★ ";
    color: gold;
}

/* After (insert content after) */
h1::after {
    content: " ★";
    color: gold;
}

/* Selection (when user selects text) */
::selection {
    background-color: yellow;
    color: black;
}

```

CHAPTER 4: COLORS IN CSS

A) COLOR NAMES (140+ named colors):

CODE:

```

p { color: red; }
p { color: blue; }
p { color: green; }
p { color: orange; }
p { color: navy; }
p { color: tomato; }
p { color: coral; }
p { color: darkcyan; }
p { color: hotpink; }
p { color: indigo; }

```

B) HEX COLORS (#RRGGBB):

CODE:

```
p { color: #FF0000; } /* Red */
p { color: #00FF00; } /* Green */
p { color: #0000FF; } /* Blue */
p { color: #000000; } /* Black */
p { color: #FFFFFF; } /* White */
p { color: #FFA500; } /* Orange */
p { color: #800080; } /* Purple */
p { color: #008080; } /* Teal */
```

/ Shorthand */*

```
p { color: #F00; } /* #FF0000 (Red) */
p { color: #0F0; } /* #00FF00 (Green) */
p { color: #00F; } /* #0000FF (Blue) */
```

C) RGB COLORS:

CODE:

```
p { color: rgb(255, 0, 0); } /* Red */
p { color: rgb(0, 255, 0); } /* Green */
p { color: rgb(0, 0, 255); } /* Blue */
p { color: rgb(255, 255, 255); } /* White */
p { color: rgb(0, 0, 0); } /* Black */
```

/ RGB with Alpha (Transparency) */*

```
p { color: rgba(255, 0, 0, 0.5); } /* 50% transparent Red */
p { background: rgba(0, 0, 255, 0.3); } /* 30% transparent Blue bg */
```

D) HSL COLORS:

CODE:

/ Hue(0-360) Saturation(0-100%) Lightness(0-100%) */*

```
p { color: hsl(0, 100%, 50%); } /* Red */
p { color: hsl(120, 100%, 50%); } /* Green */
p { color: hsl(240, 100%, 50%); } /* Blue */
p { color: hsl(0, 0%, 0%); } /* Black */
p { color: hsl(0, 0%, 100%); } /* White */
```

/ HSL with Alpha */*

```
p { color: hsla(0, 100%, 50%, 0.5); } /* 50% transparent Red */
```


E) CURRENT COLOR:

CODE:

```
div {  
    color: blue;  
    border: 2px solid currentColor; /* Blue border */  
}
```

CHAPTER 5: BACKGROUNDS

A) BACKGROUND COLOR:

CODE:

```
div {  
    background-color: lightblue;  
}
```

B) BACKGROUND IMAGE:

CODE:

```
div {  
    background-image: url("image.jpg");  
    background-size: cover; /* Cover entire area */  
    background-repeat: no-repeat; /* Don't repeat */  
    background-position: center; /* Center the image */  
}
```

C) BACKGROUND SHORTHAND:

CODE:

```
div {
```

```
background: lightblue url("image.jpg") no-repeat center / cover;
}
```

D) BACKGROUND PROPERTIES:

CODE:

```
div {
  background-color: #f0f0f0;
  background-image: url("pattern.png");
  background-repeat: repeat-x;    /* Repeat horizontally */
  background-repeat: repeat-y;    /* Repeat vertically */
  background-repeat: no-repeat;    /* No repeat */
  background-position: top left;  /* Position */
  background-position: 50% 50%;  /* Center */
  background-size: cover;        /* Cover area */
  background-size: contain;      /* Fit inside */
  background-size: 100% 100%;    /* Stretch */
  background-attachment: fixed;   /* Fixed background */
  background-attachment: scroll;   /* Scroll with page */
}
```

E) GRADIENT BACKGROUNDS:

CODE:

```
/* Linear Gradient */
div {
  background: linear-gradient(red, blue);
}
div {
  background: linear-gradient(to right, red, orange, yellow);
}
div {
  background: linear-gradient(45deg, #ff6b6b, #4ecdc4);
}

/* Radial Gradient */
div {
  background: radial-gradient(circle, red, blue);
}
div {
  background: radial-gradient(ellipse, red, yellow, blue);
}
```

```
/* Multiple Backgrounds */
div {
  background:
    linear-gradient(rgba(0,0,0,0.5), rgba(0,0,0,0.5)),
    url("image.jpg") center/cover;
}
```

CHAPTER 6: TEXT STYLING

A) FONT PROPERTIES:

CODE:

```
p {
  font-family: Arial, Helvetica, sans-serif; /* Font family */
  font-size: 16px; /* Font size (px, em, rem, %) */
  font-weight: bold; /* bold, normal, 100-900 */
  font-style: italic; /* italic, normal, oblique */
  font-variant: small-caps; /* Small caps */
  line-height: 1.6; /* Space between lines */
}
```

/* Font Shorthand */

```
p {
  font: italic bold 16px/1.6 Arial, sans-serif;
}
```

B) TEXT PROPERTIES:

CODE:

```
p {
  color: #333; /* Text color */
  text-align: center; /* left, right, center, justify */
  text-decoration: underline; /* underline, overline, line-through, none */
  text-transform: uppercase; /* uppercase, lowercase, capitalize */
  text-indent: 20px; /* First line indent */
  letter-spacing: 2px; /* Space between letters */
  word-spacing: 5px; /* Space between words */
}
```

```
text-shadow: 2px 2px 4px rgba(0,0,0,0.3); /* Shadow */
direction: ltr; /* ltr, rtl */
}
```

C) TEXT SHADOW EXAMPLES:

CODE:

```
h1 {
  text-shadow: 2px 2px 4px rgba(0,0,0,0.5);
}

h1 {
  text-shadow: 0 0 10px red; /* Glow effect */
}

h1 {
  text-shadow:
    1px 1px 0 red,
    2px 2px 0 orange,
    3px 3px 0 yellow;
}
```

D) GOOGLE FONTS (External Fonts):

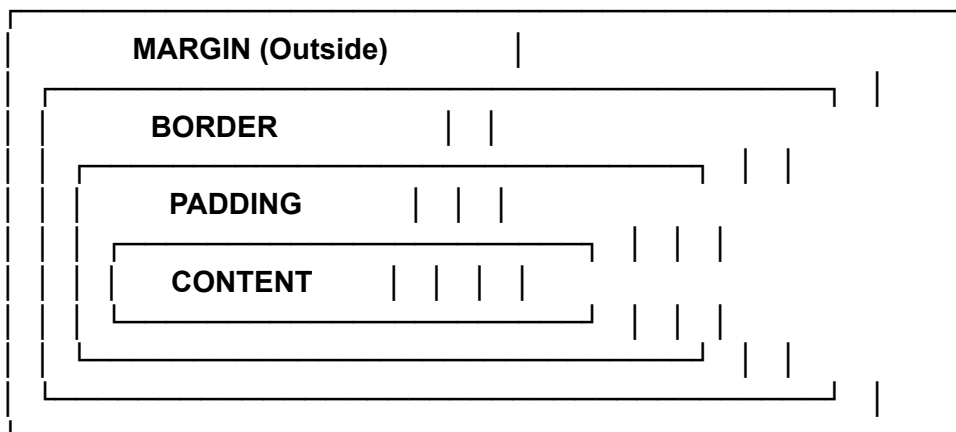
CODE:

```
<!-- In HTML <head> -->
<link
href="https://fonts.googleapis.com/css2?family=Roboto:wght@300;400;700&display=
swap" rel="stylesheet">

<!-- In CSS -->
p {
  font-family: 'Roboto', sans-serif;
  font-weight: 400;
}
```

CHAPTER 7: BOX MODEL

THE BOX MODEL (Every element is a box):



A) CONTENT:

CODE:

```
div {  
  width: 200px;  
  height: 100px;  
  max-width: 500px; /* Maximum width */  
  min-width: 100px; /* Minimum width */  
  max-height: 300px; /* Maximum height */  
  min-height: 50px; /* Minimum height */  
}
```

B) PADDING (Space inside border):

CODE:

```
div {  
  padding: 20px; /* All sides */  
  padding: 10px 20px; /* Top/Bottom, Left/Right */  
  padding: 10px 20px 30px; /* Top, L/R, Bottom */  
  padding: 10px 20px 30px 40px; /* Top, Right, Bottom, Left */  
  
  padding-top: 10px;  
  padding-right: 20px;  
  padding-bottom: 30px;  
  padding-left: 40px;  
}
```

C) BORDER:

CODE:

```
div {
  border: 2px solid black;    /* Width, Style, Color */
  border-width: 2px;
  border-style: solid;       /* solid, dotted, dashed, double, none */
  border-color: red;
  border-radius: 10px;       /* Rounded corners */

  /* Individual sides */
  border-top: 2px solid red;
  border-bottom: 4px dashed blue;
  border-left: 1px dotted green;
  border-right: 3px double orange;
}

/* Border Radius */
div {
  border-radius: 10px;       /* All corners */
  border-radius: 10px 20px 30px 40px; /* TL, TR, BR, BL */
  border-radius: 50%;       /* Circle */
  border-radius: 20px 10px;  /* TL+BR, TR+BL */
}
```

D) MARGIN (Space outside border):

CODE:

```
div {
  margin: 20px;              /* All sides */
  margin: 10px 20px;         /* Top/Bottom, Left/Right */
  margin: 10px 20px 30px;    /* Top, L/R, Bottom */
  margin: 10px 20px 30px 40px; /* Top, Right, Bottom, Left */

  margin-top: 10px;
  margin-right: 20px;
  margin-bottom: 30px;
  margin-left: 40px;

  margin: 0 auto;           /* Center horizontally */
}
```

E) BOX-SIZING (Very Important!):

CODE:

```
/* Default (content-box) - Width = content only */
div {
  box-sizing: content-box;
  width: 200px;
  padding: 20px;
  border: 2px solid black;
  /* Total width = 200 + 40 + 4 = 244px */
}

/* Border-box - Width includes padding and border */
div {
  box-sizing: border-box;
  width: 200px;
  padding: 20px;
  border: 2px solid black;
  /* Total width = 200px (includes padding and border) */
}

/* Use border-box for everything */
* {
  box-sizing: border-box;
}
```

CHAPTER 8: DISPLAY PROPERTY

A) DISPLAY VALUES:

Value	Description
block	Takes full width, new line
inline	Takes only needed space
inline-block	Inline but can set width/height
none	Element disappears (removed)
flex	Flexible box layout
grid	Grid layout

table	Like a table	
contents	Only children visible	

CODE:

/* Block - Takes full width */

```
div {
  display: block;
}
```

/* Inline - Takes only needed space */

```
span {
  display: inline;
}
```

/* Inline-block - Best of both */

```
button {
  display: inline-block;
  width: 100px; /* Works with inline-block */
  height: 40px; /* Works with inline-block */
}
```

/* None - Hidden */

```
.hidden {
  display: none; /* Element removed from flow */
}
```

/* Visibility: hidden (Hides but takes space) */

```
.invisible {
  visibility: hidden; /* Still takes space */
}
```

CHAPTER 9: FLEXBOX (Modern Layout)

A) BASIC FLEXBOX:

CODE:


```
.container {  
  display: flex; /* Enable flexbox */  
}
```

B) FLEX CONTAINER PROPERTIES:

CODE:

```
.container {  
  display: flex;  
  
  /* Direction */  
  flex-direction: row;      /* Default - left to right */  
  flex-direction: row-reverse; /* Right to left */  
  flex-direction: column;    /* Top to bottom */  
  flex-direction: column-reverse; /* Bottom to top */  
  
  /* Wrap */  
  flex-wrap: nowrap;        /* Default - no wrap */  
  flex-wrap: wrap;          /* Wrap to next line */  
  flex-wrap: wrap-reverse;  /* Wrap reverse */  
  
  /* Shorthand */  
  flex-flow: row wrap;      /* direction + wrap */  
  
  /* Justify (Main Axis) */  
  justify-content: flex-start; /* Default - start */  
  justify-content: flex-end;   /* End */  
  justify-content: center;     /* Center */  
  justify-content: space-between; /* Space between items */  
  justify-content: space-around; /* Space around items */  
  justify-content: space-evenly; /* Even space */  
  
  /* Align (Cross Axis) */  
  align-items: stretch;      /* Default - stretch */  
  align-items: flex-start;    /* Start */  
  align-items: flex-end;      /* End */  
  align-items: center;        /* Center */  
  align-items: baseline;      /* Baseline */  
  
  /* Align Multiple Rows */  
  align-content: flex-start;  /* Start */  
  align-content: flex-end;    /* End */  
  align-content: center;      /* Center */  
  align-content: space-between; /* Space between */  
  align-content: space-around; /* Space around */
```

```
align-content: stretch;    /* Stretch */

/* Gap */
gap: 20px;                  /* Space between items */
}
```

C) FLEX ITEM PROPERTIES:

CODE:

```
.item {
  /* Order */
  order: 2;          /* Default 0 - Lower number first */

  /* Grow */
  flex-grow: 1;      /* 0 = no grow, 1 = grow */
  flex-grow: 2;      /* Grows twice as much as 1 */

  /* Shrink */
  flex-shrink: 1;    /* 0 = no shrink, 1 = shrink */

  /* Basis (Initial size) */
  flex-basis: 100px; /* Starting size */
  flex-basis: auto;  /* Based on content */

  /* Shorthand */
  flex: 1;           /* grow:1 shrink:1 basis:0 */
  flex: 1 1 100px;   /* grow shrink basis */
  flex: 0 0 200px;   /* Fixed size 200px */

  /* Align Self (Individual alignment) */
  align-self: flex-start;
  align-self: flex-end;
  align-self: center;
  align-self: stretch;
}
```

D) FLEXBOX EXAMPLES:

CODE:

```
/* 1. Centering (Most common) */
.center {
  display: flex;
```

```

    justify-content: center;
    align-items: center;
    height: 100vh;
}

/* 2. Navigation Bar */
.nav {
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 20px;
    background: #2c3e50;
}

/* 3. Card Layout */
.cards {
    display: flex;
    gap: 20px;
    flex-wrap: wrap;
}
.card {
    flex: 1 1 250px;
    padding: 20px;
    border: 1px solid #ddd;
}

/* 4. Equal Height Columns */
.columns {
    display: flex;
    gap: 20px;
}
.column {
    flex: 1;
}

```

CHAPTER 10: CSS GRID (Advanced Layout)

A) BASIC GRID:

CODE:

```

.container {

```

```
display: grid;
grid-template-columns: 200px 200px 200px; /* 3 columns */
grid-template-rows: 100px 100px; /* 2 rows */
gap: 20px;
}
```

B) GRID TEMPLATE:

CODE:

```
.container {
  display: grid;

  /* Columns */
  grid-template-columns: 200px 300px 200px; /* Fixed sizes */
  grid-template-columns: 1fr 2fr 1fr; /* Fractions */
  grid-template-columns: 1fr 1fr 1fr; /* Equal 3 columns */
  grid-template-columns: repeat(3, 1fr); /* Same as above */
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr)); /* Responsive */
  grid-template-columns: minmax(100px, 300px) 1fr 1fr;

  /* Rows */
  grid-template-rows: 100px 200px;
  grid-template-rows: 1fr 2fr;
  grid-template-rows: 100px auto;

  /* Gap */
  gap: 20px; /* Both row and column */
  row-gap: 20px; /* Only rows */
  column-gap: 30px; /* Only columns */
}
```

C) GRID ITEM PLACEMENT:

CODE:

```
.item1 {
  grid-column: 1 / 3; /* Start at 1, end at 3 (span 2 columns) */
  grid-column: span 2; /* Span 2 columns */
  grid-row: 1 / 2; /* Row 1 only */

  /* Shorthand */
  grid-area: 1 / 1 / 3 / 3; /* row-start / col-start / row-end / col-end */
}
```

```
/* Named lines */
.container {
  grid-template-columns: [start] 1fr [middle] 2fr [end];
}
.item {
  grid-column: start / end;
}
```

D) GRID AREAS (Naming layout):

CODE:

```
.container {
  display: grid;
  grid-template-areas:
    "header header header"
    "sidebar main main"
    "footer footer footer";
  grid-template-columns: 200px 1fr 1fr;
  grid-template-rows: 80px 1fr 60px;
  gap: 10px;
}

.header { grid-area: header; }
.sidebar { grid-area: sidebar; }
.main { grid-area: main; }
.footer { grid-area: footer; }
```

E) GRID EXAMPLES:

CODE:

```
/* 1. Photo Gallery */
.gallery {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
  gap: 15px;
}

/* 2. Blog Layout */
.blog {
  display: grid;
  grid-template-columns: 2fr 1fr;
```

```

grid-template-rows: auto 1fr auto;
grid-template-areas:
  "header header"
  "content sidebar"
  "footer footer";
gap: 20px;
}

```

```

/* 3. Dashboard */
.dashboard {
  display: grid;
  grid-template-columns: 250px 1fr;
  grid-template-rows: 60px 1fr 50px;
  grid-template-areas:
    "header header"
    "sidebar main"
    "footer footer";
  height: 100vh;
}

```

CHAPTER 11: POSITIONING

A) POSITION VALUES:

Value	Description
static	Default - Normal flow
relative	Relative to normal position
absolute	Relative to nearest positioned ancestor
fixed	Fixed relative to viewport
sticky	Sticks when scrolling

B) POSITION EXAMPLES:

CODE:

```
/* 1. Relative (Offset from normal position) */  
.relative-box {  
    position: relative;  
    top: 20px;  
    left: 30px;  
    /* Moves 20px down, 30px right from normal */  
}
```

```
/* 2. Absolute (Positioned relative to parent) */  
.parent {  
    position: relative; /* Parent must have position */  
}  
.absolute-child {  
    position: absolute;  
    top: 0;  
    right: 0;  
    /* Top-right corner of parent */  
}
```

```
/* 3. Fixed (Stays in same place while scrolling) */  
.fixed-header {  
    position: fixed;  
    top: 0;  
    left: 0;  
    width: 100%;  
    background: white;  
    z-index: 100;  
}
```

```
/* 4. Sticky (Combination of relative and fixed) */  
.sticky-header {  
    position: sticky;  
    top: 0;  
    background: white;  
    z-index: 10;  
}
```

C) Z-INDEX (Stacking Order):

CODE:

```
/* Higher z-index appears on top */  
.box1 {  
    position: relative;  
    z-index: 1;  
}
```

```
.box2 {  
    position: relative;  
    z-index: 2; /* On top of box1 */  
}  
.box3 {  
    position: relative;  
    z-index: 3; /* On top of both */  
}
```

CHAPTER 12: FLOATS AND CLEAR

A) FLOAT:

CODE:

```
/* Float images */  
img {  
    float: left; /* Image on left, text wraps around */  
    margin-right: 20px;  
}  
img {  
    float: right; /* Image on right */  
    margin-left: 20px;  
}
```

```
/* Float columns */  
.column {  
    float: left;  
    width: 33.33%;  
    padding: 15px;  
    box-sizing: border-box;  
}
```

B) CLEAR:

CODE:

```
/* Clear both sides */  
.clearfix::after {  
    content: "";
```



```
    display: table;
    clear: both;
}

/* Clear only left or right */
.clear-left {
    clear: left;
}
.clear-right {
    clear: right;
}
.clear-both {
    clear: both;
}
```

CHAPTER 13: TRANSITIONS AND ANIMATIONS

A) TRANSITIONS (Smooth changes):

CODE:

```
/* Basic transition */
button {
    background-color: blue;
    transition: background-color 0.3s ease;
}
button:hover {
    background-color: red;
    /* Changes smoothly over 0.3 seconds */
}
```

```
/* Multiple properties */
.box {
    width: 100px;
    height: 100px;
    background: red;
    transition:
        width 0.5s ease,
        height 0.5s ease,
        background 0.3s ease;
}
.box:hover {
```

```

width: 200px;
height: 200px;
background: blue;
}

/* Shorthand */
div {
  transition: all 0.3s ease-in-out;
}

/* Transition Properties */
div {
  transition-property: all;      /* Which properties */
  transition-duration: 0.5s;    /* How long */
  transition-timing-function: ease; /* ease, linear, ease-in, ease-out */
  transition-delay: 0.2s;      /* Delay before starting */
}

```

B) ANIMATIONS (Keyframes):

CODE:

```

/* 1. Define keyframes */
@keyframes slide {
  0% { transform: translateX(-100%); }
  100% { transform: translateX(0); }
}

@keyframes bounce {
  0%, 100% { transform: translateY(0); }
  50% { transform: translateY(-50px); }
}

@keyframes rotate {
  0% { transform: rotate(0deg); }
  100% { transform: rotate(360deg); }
}

@keyframes fadeIn {
  0% { opacity: 0; }
  100% { opacity: 1; }
}

/* 2. Apply animation */
.element {
  animation: slide 1s ease forwards;
}

```

```
}
```

```
/* Animation Properties */
```

```
.element {  
  animation-name: slide;      /* Name of keyframes */  
  animation-duration: 2s;     /* How long */  
  animation-timing-function: ease-in-out; /* ease, linear, etc. */  
  animation-delay: 0.5s;      /* Delay */  
  animation-iteration-count: infinite; /* Number of times */  
  animation-direction: alternate; /* normal, reverse, alternate */  
  animation-fill-mode: forwards; /* Stay at end state */  
  animation-play-state: paused; /* paused, running */  
}
```

```
/* Shorthand */
```

```
.element {  
  animation: slide 2s ease-in-out 0.5s infinite alternate;  
}
```

C) TRANSFORM (Rotate, Scale, Translate):

CODE:

```
div {  
  /* Translate (move) */  
  transform: translate(50px, 100px); /* Move X and Y */  
  transform: translateX(50px);      /* Move X only */  
  transform: translateY(100px);     /* Move Y only */  
  
  /* Rotate */  
  transform: rotate(45deg);          /* Rotate 45 degrees */  
  transform: rotateX(30deg);         /* Rotate X axis */  
  transform: rotateY(30deg);         /* Rotate Y axis */  
  transform: rotateZ(45deg);         /* Rotate Z axis */  
  
  /* Scale (size) */  
  transform: scale(1.5);              /* 1.5x size */  
  transform: scaleX(2);               /* Double width */  
  transform: scaleY(2);               /* Double height */  
  
  /* Skew */  
  transform: skew(20deg);             /* Skew */  
  transform: skewX(20deg);  
  transform: skewY(20deg);  
  
  /* Multiple transforms */
```

```
    transform: translate(50px, 50px) rotate(45deg) scale(1.5);
}
```

```
/* 3D Transforms */
```

```
div {
    transform: perspective(500px) rotateX(30deg) rotateY(30deg);
}
```

CHAPTER 14: RESPONSIVE DESIGN (Media Queries)

A) MEDIA QUERY SYNTAX:

CODE:

```
/* For screens up to 768px (Mobile) */
```

```
@media (max-width: 768px) {
    body {
        font-size: 14px;
    }
    .container {
        padding: 10px;
    }
}
```

```
/* For screens from 769px to 1024px (Tablet) */
```

```
@media (min-width: 769px) and (max-width: 1024px) {
    body {
        font-size: 16px;
    }
}
```

```
/* For screens above 1024px (Desktop) */
```

```
@media (min-width: 1025px) {
    body {
        font-size: 18px;
    }
}
```

B) COMMON BREAKPOINTS:

CODE:

/* Mobile First - Start with mobile then expand */

```
.container {  
  padding: 10px;  
}
```

/* Tablet */

```
@media (min-width: 768px) {  
  .container {  
    padding: 20px;  
    max-width: 720px;  
    margin: 0 auto;  
  }  
}
```

/* Desktop */

```
@media (min-width: 1024px) {  
  .container {  
    padding: 30px;  
    max-width: 960px;  
  }  
}
```

/* Large Desktop */

```
@media (min-width: 1440px) {  
  .container {  
    max-width: 1200px;  
  }  
}
```

C) MEDIA TYPES:

CODE:

/* Screen (default) */

```
@media screen {  
  body { background: white; }  
}
```

/* Print - For printing */

```
@media print {  
  body { background: white; color: black; }  
  .no-print { display: none; }  
}
```

```
/* All devices */
@media all {
  /* Applies to all */
}
```

D) ORIENTATION:

CODE:

```
/* Portrait mode (height > width) */
@media (orientation: portrait) {
  .sidebar { display: none; }
}
```

```
/* Landscape mode (width > height) */
@media (orientation: landscape) {
  .sidebar { display: block; }
}
```

E) RESPONSIVE DESIGN PATTERNS:

CODE:

```
/* 1. Responsive Flexbox */
.container {
  display: flex;
  flex-wrap: wrap;
}
.item {
  flex: 1 1 250px; /* Minimum 250px, grows */
}
```

```
/* 2. Responsive Grid */
.grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 20px;
}
```

```
/* 3. Hide on mobile */
@media (max-width: 768px) {
  .desktop-only { display: none; }
}
```

```
@media (min-width: 769px) {  
  .mobile-only { display: none; }  
}
```

```
/* 4. Responsive Typography */
```

```
@media (max-width: 768px) {  
  h1 { font-size: 24px; }  
  p { font-size: 14px; }  
}
```

```
@media (min-width: 769px) {  
  h1 { font-size: 36px; }  
  p { font-size: 18px; }  
}
```

CHAPTER 15: COMPLETE PROJECT - MODERN PORTFOLIO

HTML FILE (index.html):

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Krutik - Cyber Expert</title>  
  <link rel="stylesheet" href="style.css">  
  <link  
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;700&dis  
play=swap" rel="stylesheet">  
</head>  
<body>  
  <!-- Navigation -->  
  <nav class="navbar">  
    <div class="container">  
      <div class="logo">  KRUTIK</div>  
      <ul class="nav-links">  
        <li><a href="#home">Home</a></li>  
        <li><a href="#about">About</a></li>  
        <li><a href="#skills">Skills</a></li>  
        <li><a href="#projects">Projects</a></li>  
        <li><a href="#contact">Contact</a></li>  
      </ul>  
      <div class="hamburger">☰</div>
```

```
</div>
</nav>
```

```
<!-- Hero Section -->
```

```
<section id="home" class="hero">
```

```
<div class="container">
```

```
<div class="hero-content">
```

```
<h1>I'm <span class="highlight">Krutik</span></h1>
```

```
<h2>Cyber Security Expert & Developer</h2>
```

```
<p>Securing the digital world, one line of code at a time.</p>
```

```
<div class="hero-buttons">
```

```
<a href="#projects" class="btn btn-primary">View My Work</a>
```

```
<a href="#contact" class="btn btn-secondary">Contact Me</a>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- About Section -->
```

```
<section id="about" class="about">
```

```
<div class="container">
```

```
<h2 class="section-title">About Me</h2>
```

```
<div class="about-content">
```

```
<div class="about-text">
```

```
<p>I'm a passionate Cyber Security Expert with 5+ years of experience in
ethical hacking, web security, and Python development.</p>
```

```
<p>I believe in making the internet a safer place by building secure
applications and teaching others to code safely.</p>
```

```
<div class="stats">
```

```
<div class="stat">
```

```
<span class="stat-number">5+</span>
```

```
<span class="stat-label">Years Experience</span>
```

```
</div>
```

```
<div class="stat">
```

```
<span class="stat-number">50+</span>
```

```
<span class="stat-label">Projects</span>
```

```
</div>
```

```
<div class="stat">
```

```
<span class="stat-number">1000+</span>
```

```
<span class="stat-label">Students</span>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="about-image">
```

```

```

```
</div>
```

```
</div>
```

```
</div>
```


</section>

<!-- Skills Section -->

```
<section id="skills" class="skills">
  <div class="container">
    <h2 class="section-title">My Skills</h2>
    <div class="skills-grid">
      <div class="skill-item">
        <div class="skill-icon">🐍</div>
        <h3>Python</h3>
        <div class="skill-bar"><div class="skill-progress"
style="width:90%"></div></div>
      </div>
      <div class="skill-item">
        <div class="skill-icon">🌐</div>
        <h3>Web Development</h3>
        <div class="skill-bar"><div class="skill-progress"
style="width:85%"></div></div>
      </div>
      <div class="skill-item">
        <div class="skill-icon">🔒</div>
        <h3>Cyber Security</h3>
        <div class="skill-bar"><div class="skill-progress"
style="width:95%"></div></div>
      </div>
      <div class="skill-item">
        <div class="skill-icon">🗄️</div>
        <h3>SQL</h3>
        <div class="skill-bar"><div class="skill-progress"
style="width:80%"></div></div>
      </div>
      <div class="skill-item">
        <div class="skill-icon">🐧</div>
        <h3>Linux</h3>
        <div class="skill-bar"><div class="skill-progress"
style="width:85%"></div></div>
      </div>
      <div class="skill-item">
        <div class="skill-icon">📱</div>
        <h3>Responsive Design</h3>
        <div class="skill-bar"><div class="skill-progress"
style="width:90%"></div></div>
      </div>
    </div>
  </div>
</section>
```

<!-- Projects Section -->

```

<section id="projects" class="projects">
  <div class="container">
    <h2 class="section-title">My Projects</h2>
    <div class="projects-grid">
      <div class="project-card">
        <div class="project-image">🔒</div>
        <h3>Secure Login System</h3>
        <p>Authentication with password hashing, JWT tokens, and 2FA.</p>
        <a href="#" class="btn btn-small">View Project</a>
      </div>
      <div class="project-card">
        <div class="project-image">🌐</div>
        <h3>Portfolio Website</h3>
        <p>Modern responsive portfolio built with HTML5 and CSS3.</p>
        <a href="#" class="btn btn-small">View Project</a>
      </div>
      <div class="project-card">
        <div class="project-image">🛡️</div>
        <h3>Network Scanner</h3>
        <p>Python tool for network security scanning and vulnerability
detection.</p>
        <a href="#" class="btn btn-small">View Project</a>
      </div>
    </div>
  </div>
</section>

<!-- Contact Section -->
<section id="contact" class="contact">
  <div class="container">
    <h2 class="section-title">Contact Me</h2>
    <div class="contact-content">
      <form class="contact-form">
        <input type="text" placeholder="Your Name" required>
        <input type="email" placeholder="Your Email" required>
        <textarea rows="5" placeholder="Your Message" required></textarea>
        <button type="submit" class="btn btn-primary">Send Message</button>
      </form>
    </div>
  </div>
</section>

<!-- Footer -->
<footer class="footer">
  <div class="container">
    <p>&copy; 2026 Krutik 🌟 Cyber Expert. All Rights Reserved.</p>
    <div class="social-links">
      <a href="#">LinkedIn</a>
    </div>
  </div>
</footer>

```

```
        <a href="#">GitHub</a>
        <a href="#">Twitter</a>
        <a href="#">YouTube</a>
    </div>
</div>
</footer>
</body>
</html>
```

CSS FILE (style.css):

```
/* ===== RESET ===== */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Poppins', sans-serif;
    line-height: 1.6;
    color: #333;
}

.container {
    max-width: 1200px;
    margin: 0 auto;
    padding: 0 20px;
}

/* ===== NAVIGATION ===== */
.navbar {
    background: #2c3e50;
    color: white;
    padding: 15px 0;
    position: fixed;
    width: 100%;
    top: 0;
    z-index: 1000;
}

.navbar .container {
    display: flex;
    justify-content: space-between;
    align-items: center;
}
```

```
.logo {
  font-size: 24px;
  font-weight: 700;
}

.nav-links {
  display: flex;
  list-style: none;
  gap: 30px;
}

.nav-links a {
  color: white;
  text-decoration: none;
  transition: color 0.3s;
}

.nav-links a:hover {
  color: #f1c40f;
}

.hamburger {
  display: none;
  font-size: 28px;
  cursor: pointer;
}

/* ===== HERO ===== */
.hero {
  background: linear-gradient(135deg, #2c3e50, #4ca1af);
  color: white;
  padding: 150px 0 100px;
  min-height: 100vh;
  display: flex;
  align-items: center;
}

.hero-content h1 {
  font-size: 48px;
  margin-bottom: 10px;
}

.hero-content h2 {
  font-size: 24px;
  font-weight: 400;
  margin-bottom: 20px;
}
```

```
.highlight {  
  color: #f1c40f;  
}
```

```
.hero-content p {  
  font-size: 18px;  
  margin-bottom: 30px;  
  opacity: 0.9;  
}
```

```
.hero-buttons {  
  display: flex;  
  gap: 20px;  
}
```

```
.btn {  
  display: inline-block;  
  padding: 12px 30px;  
  border-radius: 30px;  
  text-decoration: none;  
  font-weight: 600;  
  transition: all 0.3s;  
}
```

```
.btn-primary {  
  background: #f1c40f;  
  color: #2c3e50;  
}
```

```
.btn-primary:hover {  
  background: #f39c12;  
  transform: translateY(-2px);  
}
```

```
.btn-secondary {  
  background: transparent;  
  color: white;  
  border: 2px solid white;  
}
```

```
.btn-secondary:hover {  
  background: white;  
  color: #2c3e50;  
}
```

```
.btn-small {  
  padding: 8px 20px;
```

```
    font-size: 14px;
}

/* ===== SECTIONS ===== */
section {
    padding: 80px 0;
}

.section-title {
    font-size: 36px;
    text-align: center;
    margin-bottom: 50px;
}

.section-title::after {
    content: "";
    display: block;
    width: 60px;
    height: 4px;
    background: #f1c40f;
    margin: 10px auto 0;
}

/* ===== ABOUT ===== */
.about {
    background: #f8f9fa;
}

.about-content {
    display: flex;
    gap: 50px;
    align-items: center;
}

.about-text {
    flex: 1;
}

.about-text p {
    margin-bottom: 15px;
}

.stats {
    display: flex;
    gap: 40px;
    margin-top: 30px;
}
```

```
.stat {
  text-align: center;
}

.stat-number {
  display: block;
  font-size: 32px;
  font-weight: 700;
  color: #2c3e50;
}

.stat-label {
  font-size: 14px;
  color: #666;
}

.about-image {
  flex: 1;
  text-align: center;
}

.about-image img {
  max-width: 100%;
  border-radius: 10px;
}

/* ===== SKILLS ===== */
.skills-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
  gap: 30px;
}

.skill-item {
  background: white;
  padding: 30px;
  border-radius: 10px;
  box-shadow: 0 2px 10px rgba(0,0,0,0.1);
  text-align: center;
}

.skill-icon {
  font-size: 40px;
  margin-bottom: 10px;
}

.skill-item h3 {
  margin-bottom: 15px;
}
```

```
}
```

```
.skill-bar {  
  width: 100%;  
  height: 10px;  
  background: #e0e0e0;  
  border-radius: 5px;  
  overflow: hidden;  
}
```

```
.skill-progress {  
  height: 100%;  
  background: #f1c40f;  
  border-radius: 5px;  
  transition: width 1s ease;  
}
```

```
/* ===== PROJECTS ===== */
```

```
.projects {  
  background: #f8f9fa;  
}
```

```
.projects-grid {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));  
  gap: 30px;  
}
```

```
.project-card {  
  background: white;  
  padding: 30px;  
  border-radius: 10px;  
  box-shadow: 0 2px 10px rgba(0,0,0,0.1);  
  text-align: center;  
  transition: transform 0.3s;  
}
```

```
.project-card:hover {  
  transform: translateY(-10px);  
}
```

```
.project-image {  
  font-size: 60px;  
  margin-bottom: 15px;  
}
```

```
.project-card h3 {  
  margin-bottom: 10px;
```



```

}

.project-card p {
    margin-bottom: 15px;
    color: #666;
}

/* ===== CONTACT ===== */
.contact-form {
    max-width: 600px;
    margin: 0 auto;
}

.contact-form input,
.contact-form textarea {
    width: 100%;
    padding: 15px;
    margin-bottom: 20px;
    border: 2px solid #ddd;
    border-radius: 8px;
    font-family: 'Poppins', sans-serif;
    transition: border-color 0.3s;
}

.contact-form input:focus,
.contact-form textarea:focus {
    outline: none;
    border-color: #f1c40f;
}

.contact-form .btn {
    width: 100%;
    border: none;
    cursor: pointer;
}

/* ===== FOOTER ===== */
.footer {
    background: #2c3e50;
    color: white;
    text-align: center;
    padding: 30px 0;
}

.footer p {
    margin-bottom: 15px;
}

```

```

.social-links {
    display: flex;
    justify-content: center;
    gap: 20px;
}

.social-links a {
    color: white;
    text-decoration: none;
    transition: color 0.3s;
}

.social-links a:hover {
    color: #f1c40f;
}

/* ===== RESPONSIVE ===== */

/* Tablet */
@media (max-width: 992px) {
    .about-content {
        flex-direction: column;
        text-align: center;
    }

    .stats {
        justify-content: center;
    }
}

/* Mobile */
@media (max-width: 768px) {
    .nav-links {
        display: none;
        flex-direction: column;
        width: 100%;
        text-align: center;
        padding: 20px 0;
        background: #2c3e50;
    }

    .nav-links.active {
        display: flex;
    }

    .hamburger {
        display: block;
    }
}

```

```
.hero-content h1 {  
  font-size: 32px;  
}  
  
.hero-content h2 {  
  font-size: 18px;  
}  
  
.hero-buttons {  
  flex-direction: column;  
  align-items: center;  
}  
  
.section-title {  
  font-size: 28px;  
}  
  
.stats {  
  flex-direction: column;  
  gap: 15px;  
}  
  
.projects-grid,  
.skills-grid {  
  grid-template-columns: 1fr;  
}  
}
```

CHAPTER 16: QUICK REFERENCE - ALL CSS PROPERTIES

FONT PROPERTIES:

font-family, font-size, font-weight, font-style, font-variant, line-height, font

TEXT PROPERTIES:

color, text-align, text-decoration, text-transform, text-indent, text-shadow,
letter-spacing, word-spacing

BACKGROUND PROPERTIES:

background-color, background-image, background-repeat, background-position,
background-size, background-attachment, background

BOX MODEL:

width, height, max-width, min-width, max-height, min-height, padding, margin, border, border-radius, box-shadow, box-sizing

DISPLAY & POSITION:

display, position, top, right, bottom, left, z-index, float, clear, visibility, opacity

FLEXBOX:

display:flex, flex-direction, flex-wrap, flex-flow, justify-content, align-items, align-content, gap, flex-grow, flex-shrink, flex-basis, flex, order, align-self

GRID:

display:grid, grid-template-columns, grid-template-rows, grid-template-areas, gap, row-gap, column-gap, grid-column, grid-row, grid-area

ANIMATIONS:

transition, animation, @keyframes, transform

RESPONSIVE:

@media (max-width: 768px) { ... }

CERTIFICATE OF COMPLETION

This certifies that the reader has completed

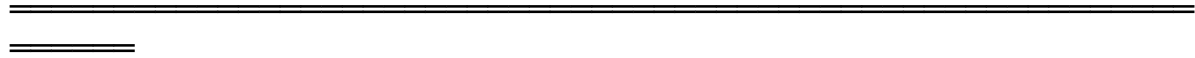
CSS COMPLETE COURSE

By
KRUTIK  CYBER EXPERT

Topics Covered:

- ✓ CSS Introduction (3 Ways to Add CSS)
- ✓ CSS Selectors (All Types)
- ✓ Colors (Named, HEX, RGB, HSL)
- ✓ Backgrounds (Colors, Images, Gradients)
- ✓ Text Styling (Fonts, Shadows, Google Fonts)
- ✓ Box Model (Content, Padding, Border, Margin)
- ✓ Display Property
- ✓ Flexbox (Modern Layout)
- ✓ CSS Grid (Advanced Layout)
- ✓ Positioning (Static, Relative, Absolute, Fixed, Sticky)
- ✓ Floats and Clear
- ✓ Transitions and Animations
- ✓ Responsive Design (Media Queries)
- ✓ Complete Project (Modern Portfolio)

"Make it Beautiful."



**NOTE: This PDF is completely FREE. If you paid for it, you got scammed.
Share it with everyone who wants to learn CSS! 🚀**

